

| LLM           | Method                        | HuggingFace            |                        |                       | Multimedia             |                        |                       | Daily Life             |                        |                       |
|---------------|-------------------------------|------------------------|------------------------|-----------------------|------------------------|------------------------|-----------------------|------------------------|------------------------|-----------------------|
|               |                               | <i>n-F1</i> $\uparrow$ | <i>l-F1</i> $\uparrow$ | <i>Acc</i> $\uparrow$ | <i>n-F1</i> $\uparrow$ | <i>l-F1</i> $\uparrow$ | <i>Acc</i> $\uparrow$ | <i>n-F1</i> $\uparrow$ | <i>l-F1</i> $\uparrow$ | <i>Acc</i> $\uparrow$ |
| Vicuna-13B    | Direct                        | 50.46                  | 21.27                  | 8.72                  | 53.57                  | 23.19                  | 11.20                 | 73.70                  | 45.80                  | 24.43                 |
|               | TAPE                          | 59.47                  | NA                     | 5.07                  | 54.97                  | NA                     | 2.07                  | 73.26                  | NA                     | 12.50                 |
|               | GraphToken                    | <b>63.37</b>           | 31.54                  | 15.61                 | 65.40                  | 36.38                  | 19.87                 | 81.65                  | 48.29                  | 43.37                 |
|               | GraphSAGE                     | 61.86                  | 35.68                  | <b>20.08</b>          | 63.71                  | 39.88                  | 21.37                 | <b>86.07</b>           | <b>67.63</b>           | <b>48.64</b>          |
|               | GraphSAGE <sub>co-train</sub> | 62.82                  | <b>37.04</b>           | 19.68                 | <b>65.89</b>           | <b>42.18</b>           | <b>21.58</b>          | 84.23                  | 65.44                  | 47.81                 |
| Mistral-7B    | Direct                        | 60.60                  | 30.23                  | 16.36                 | 69.83                  | 39.85                  | 25.05                 | 84.26                  | 53.63                  | 44.52                 |
|               | TAPE                          | 61.82                  | NA                     | 6.13                  | 58.92                  | NA                     | 3.29                  | 76.40                  | NA                     | 16.44                 |
|               | GraphToken                    | 64.42                  | 32.04                  | 18.60                 | 72.31                  | 42.60                  | 30.31                 | 86.82                  | 57.06                  | <b>53.99</b>          |
|               | GraphSAGE                     | <b>68.12</b>           | 43.09                  | 25.77                 | 75.51                  | 52.94                  | <b>34.29</b>          | 87.51                  | 66.57                  | 52.74                 |
|               | GraphSAGE <sub>co-train</sub> | 67.61                  | <b>43.14</b>           | <b>27.20</b>          | <b>76.96</b>           | <b>55.46</b>           | 33.26                 | <b>87.61</b>           | <b>66.75</b>           | 52.97                 |
| CodeLlama-13B | Direct                        | 57.55                  | 28.88                  | 14.29                 | 68.57                  | 41.79                  | 24.10                 | 91.20                  | 76.07                  | 66.40                 |
|               | TAPE                          | 64.03                  | NA                     | 8.05                  | 58.27                  | NA                     | 2.01                  | 77.74                  | NA                     | 17.37                 |
|               | GraphToken                    | 62.15                  | 32.55                  | 20.08                 | 74.57                  | 47.60                  | 35.06                 | 92.50                  | 73.57                  | 69.42                 |
|               | GraphSAGE                     | 67.30                  | 42.41                  | 26.56                 | 74.93                  | 54.52                  | <b>38.55</b>          | <b>93.84</b>           | <b>80.38</b>           | 73.60                 |
|               | GraphSAGE <sub>co-train</sub> | <b>68.92</b>           | <b>44.85</b>           | <b>29.58</b>          | <b>76.28</b>           | <b>55.41</b>           | 37.75                 | 93.30                  | 79.51                  | <b>74.00</b>          |
| GPT-3.5-turbo | Direct                        | 73.85                  | 45.73                  | 28.95                 | 82.85                  | 62.07                  | 47.96                 | 96.09                  | 83.65                  | 81.30                 |
|               | TAPE                          | 68.00                  | NA                     | 8.83                  | 62.43                  | NA                     | 3.87                  | 70.67                  | NA                     | 8.92                  |
|               | GraphSAGE                     | <b>77.90</b>           | 52.68                  | 35.11                 | 85.29                  | 65.80                  | 53.55                 | <b>96.43</b>           | <b>86.26</b>           | <b>83.13</b>          |
|               | GraphSAGE <sub>co-train</sub> | 77.87                  | <b>53.04</b>           | <b>35.32</b>          | <b>85.51</b>           | <b>66.56</b>           | <b>55.91</b>          | 96.34                  | 86.09                  | 83.13                 |
| GPT-4-turbo   | Direct                        | 77.60                  | 52.18                  | 33.68                 | 88.29                  | 69.38                  | 60.56                 | 97.36                  | 84.58                  | 86.77                 |
|               | TAPE                          | 68.82                  | NA                     | 9.50                  | 63.94                  | NA                     | 4.02                  | 71.51                  | NA                     | 9.40                  |
|               | GraphSAGE                     | <b>78.76</b>           | 52.53                  | <b>34.09</b>          | 88.63                  | 69.65                  | 60.36                 | 97.34                  | 85.67                  | <b>86.97</b>          |
|               | GraphSAGE <sub>co-train</sub> | 78.49                  | <b>52.62</b>           | 33.88                 | <b>88.86</b>           | <b>70.25</b>           | <b>62.37</b>          | <b>97.42</b>           | <b>85.80</b>           | 86.57                 |